

Antibody Structure and Function

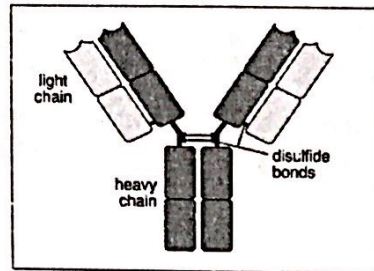


Fig 3.2 © 2001 Garland Science

Outline

- a. Describe the basic structure of an immunoglobulin.
- c. Explain electrophoresis patterns.
- d. Define hinge region.
- d. Describe the difference between polyclonal and monoclonal.
- f. Differentiate the five immunoglobulin classes found in humans.
- g. Brief description of the antibody classes and their biological functions
- h. Describe the kinetics of Ab responses following immunization

Nobel prize winner, 1972

Structure of antibody



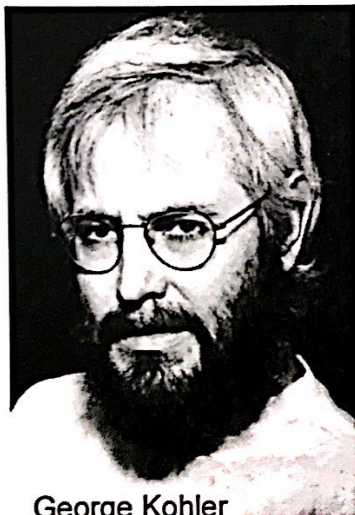
Gerald Edelman
(USA)



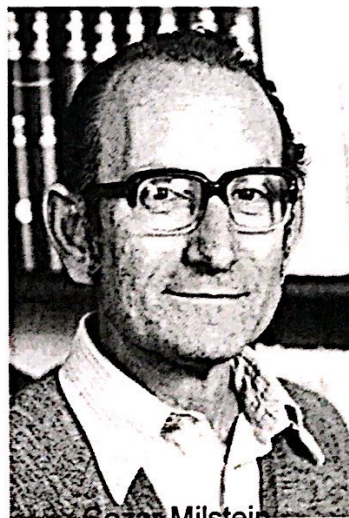
Rodney porter
(England)

Noble prize winner, 1984

Monoclonal antibodies

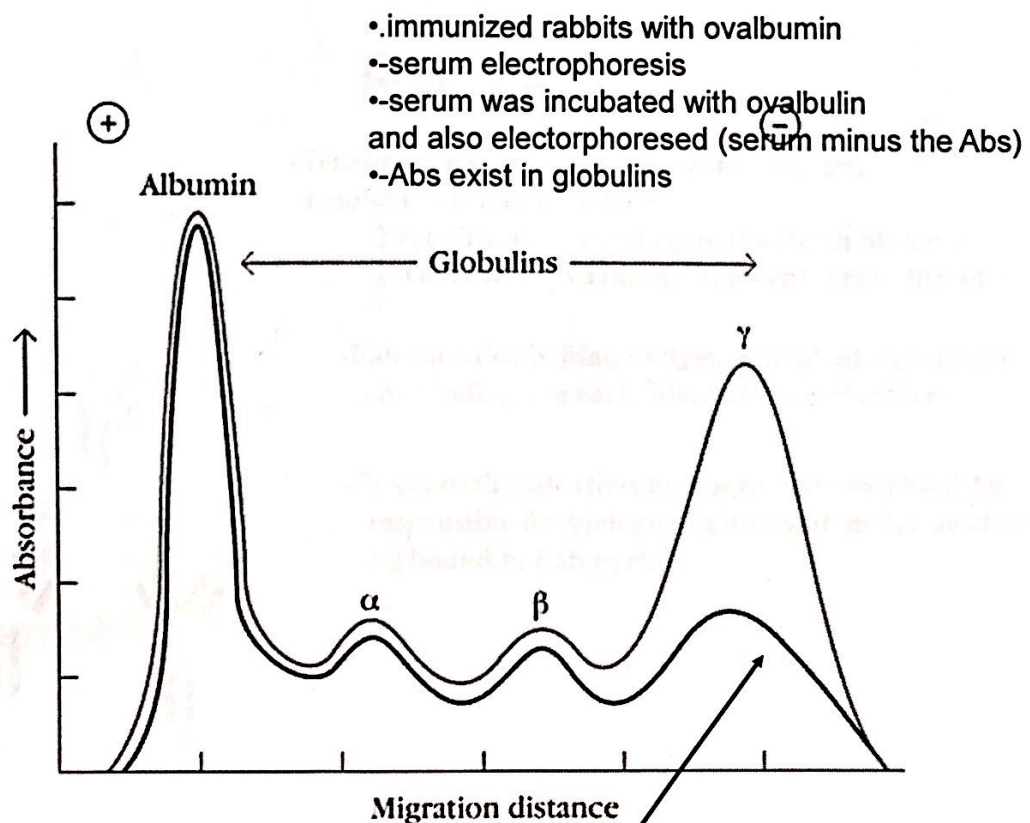
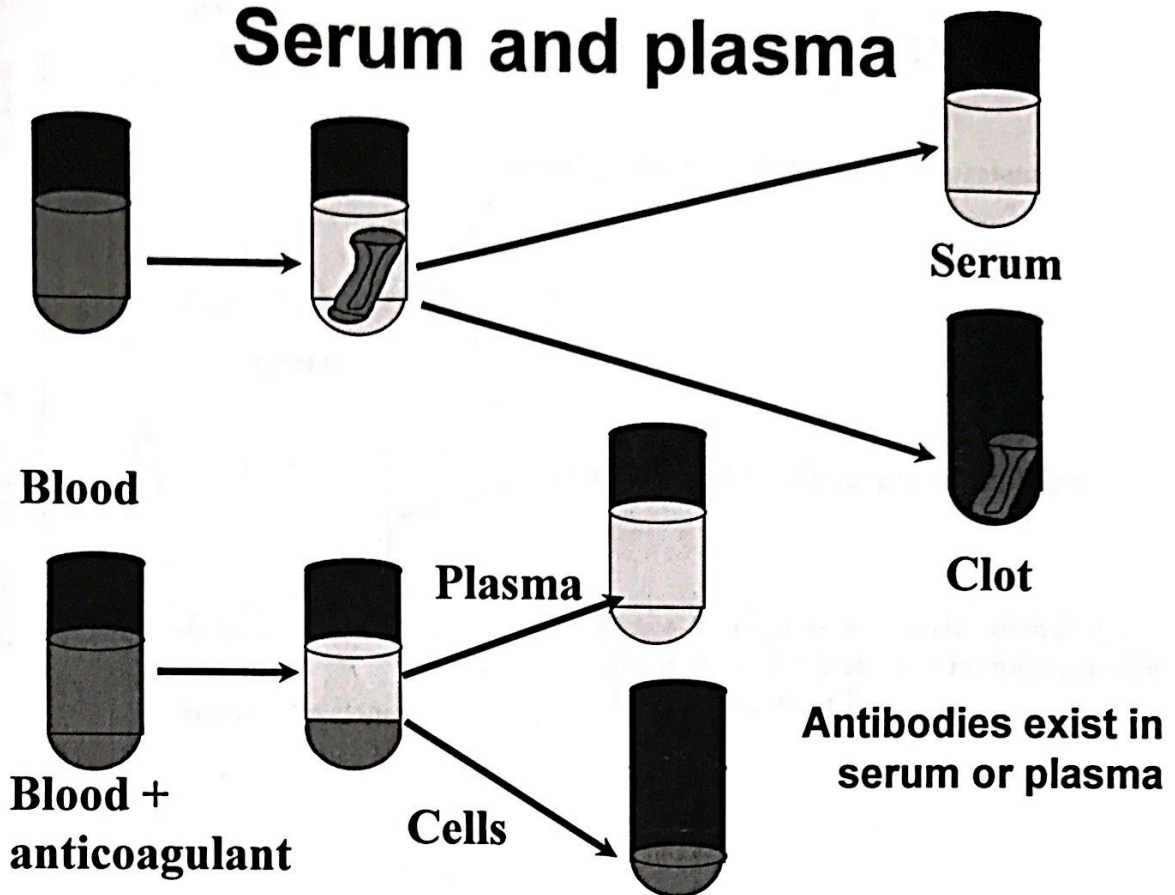


George Kohler
Germany



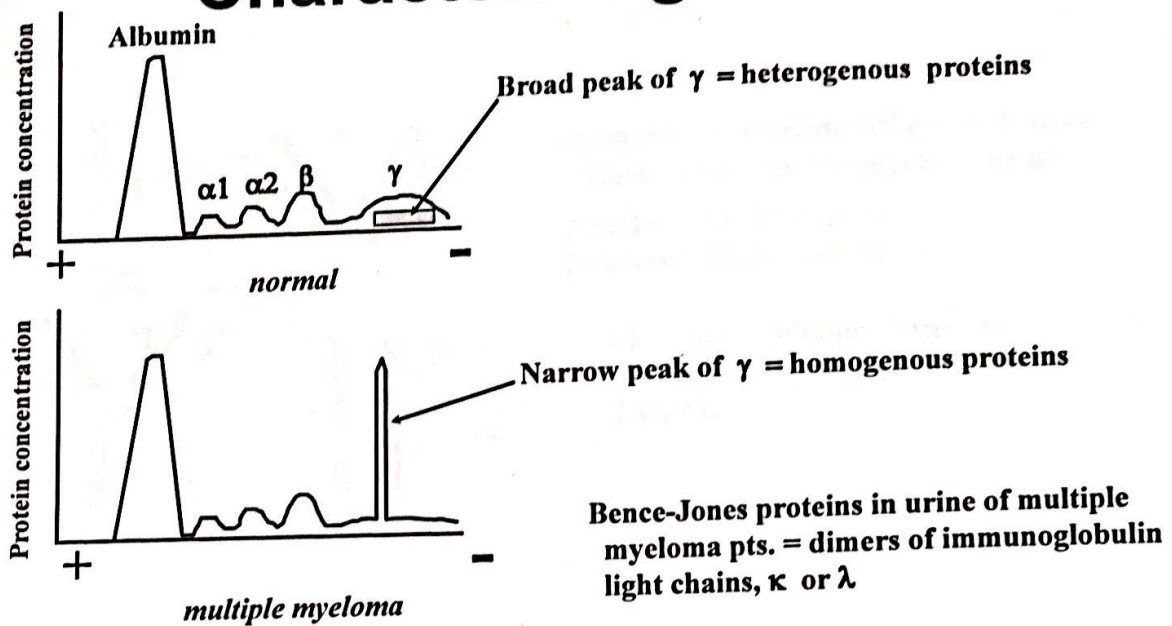
Cesar Milstein
Germany

Serum and plasma



Treated with ovalbumin (to absorb antibody)

Characterizing chains



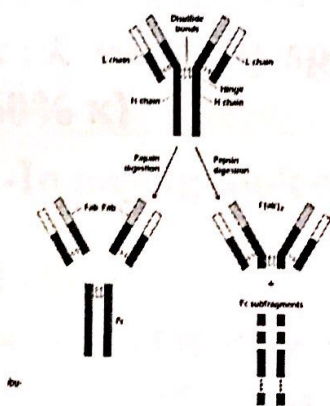
Porter

-Treatment with proteolytic enzyme, *papain*, resulted in three fragments:

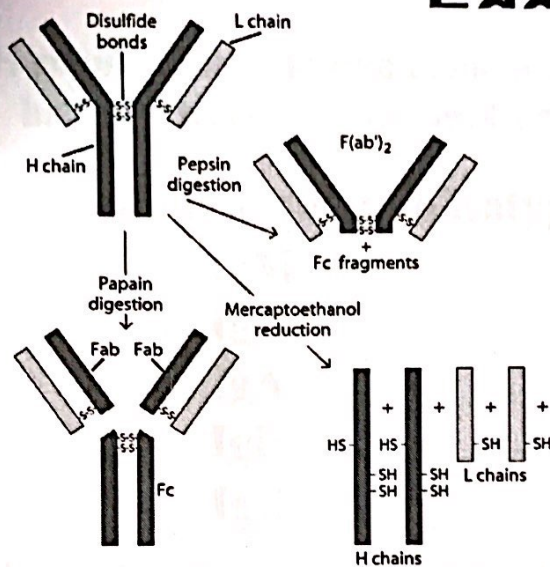
- 2 capable of Ag rx (fragment antigen binding)
- 1 could be crystallized (fragment- crystallizable)

-Fab specifically bind antigen, univalent (can't ppt) one binding site each, identical to each other

-Fc crystallizable (thus homogenous) can't bind Ag responsible for biological activity of molecule after Ag bound to Fab portions



Edelman



Treatment with mercaptoethanol = 4 chains:
(mercaptoethanol breaks S-S bonds)

2 chains = 53,000 daltons

2 chains = 22,000 daltons

All immunoglobulins basic unit consisting of 4 polypeptides i.e. 2 H, 2 L

Immunologic analysis of immunoglobulins

L chain

- All species have two major classes of L chains i.e. κ , λ
- An individual of species produces both types; ratio of $\kappa : \lambda$ varies by species (e.g. mouse 95% κ ; human 60% κ)
- In any Ig molecule, both L chains = either κ or λ

H chain:

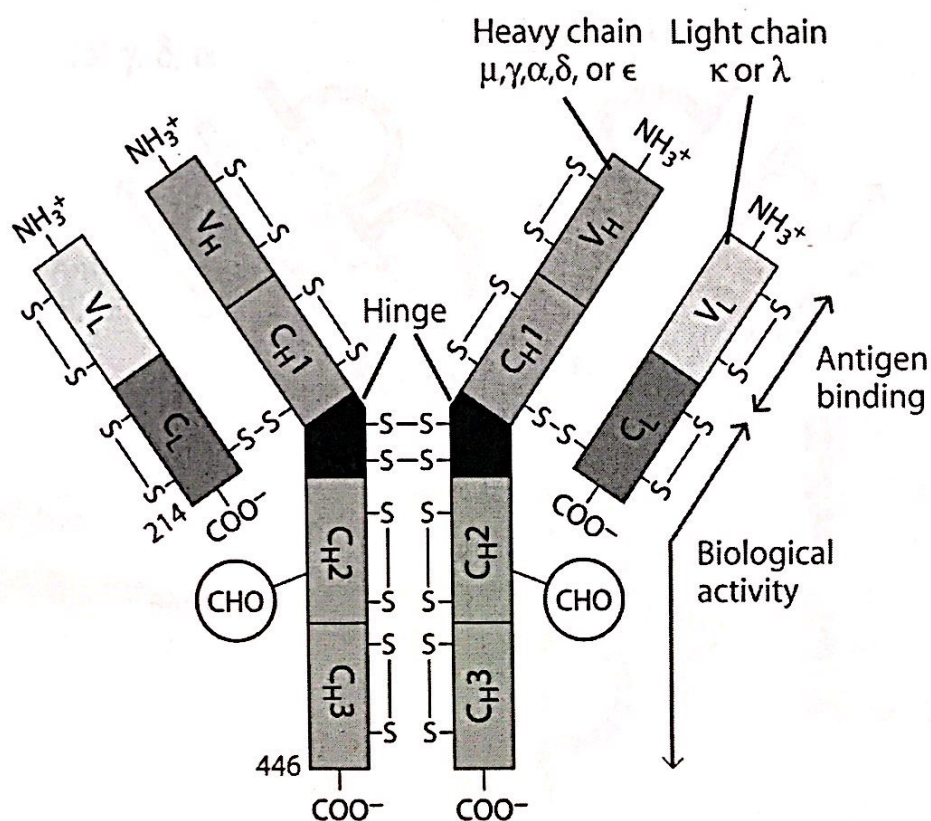
- 5 types of H chain exist in all species
- Ig consist of 5 classes (*isotypes*) differ in structure of H chains
- H chains among isotypes differ serologically, CHO content, size & biological function

H chain confers unique biologic properties of molecule e.g. half life, receptor binding, Complement activation

Immunoglobulin isotype	Heavy chain
IgM	μ
IgG	γ
IgA	α
IgD	δ
IgE	ϵ

-Individual of species produces all H chains, in proportion characteristic for species

-Ab molecule H chains are identical (i.e., 2 ϵ , 2 δ etc.)



Ig domains

- **Ig domains:** Intrachain S-S bonds form loops within the chain that create the anti-parallel β -pleated sheet
- Each L chain has 2 domains
each H chain has 4 or 5 domains
- VL, VH and CL, CH1, CH2, CH3 and CH4
- VH-VL domain
CH1-CL, CH2-CH2, CH3-CH3 domains

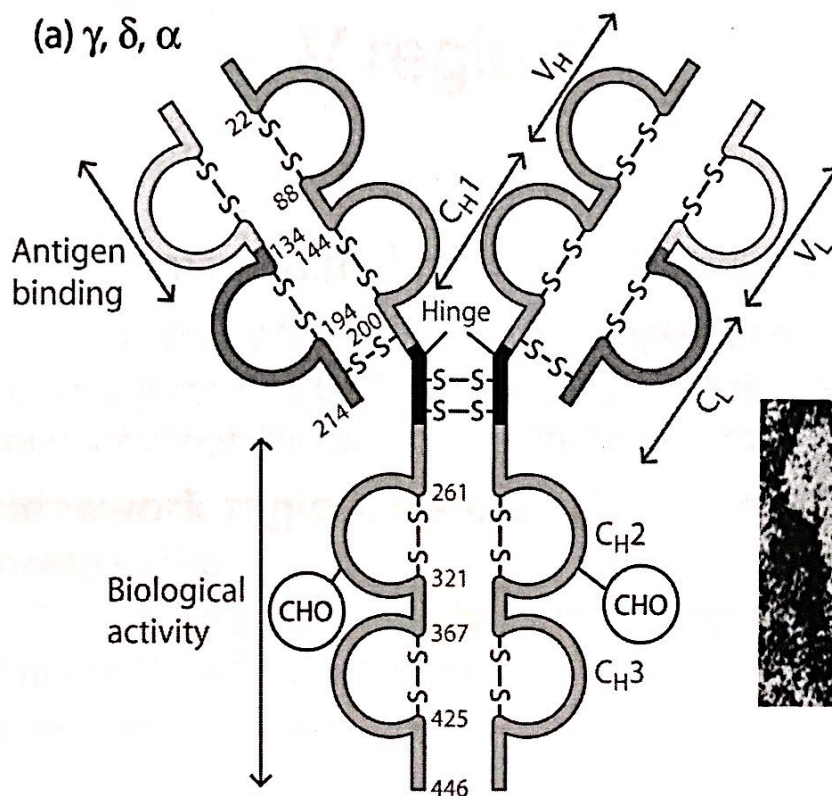


TABLE 4-1

Chain composition of the five immunoglobulin classes in humans

Class	Heavy chain	Subclasses	Light chain	Molecular formula
IgG	γ	$\gamma 1, \gamma 2, \gamma 3, \gamma 4$	κ or λ	$\gamma_2\kappa_2$ $\gamma_2\lambda_2$
IgM	μ	None	κ or λ	$(\mu_2\kappa_2)_n$ $(\mu_2\lambda_2)_n$ $n = 1 \text{ or } 5$
IgA	α	$\alpha 1, \alpha 2$	κ or λ	$(\alpha_2\kappa_2)_n$ $(\alpha_2\lambda_2)_n$ $n = 1, 2, 3, \text{ or } 4$
IgE	ϵ	None	κ or λ	$\epsilon_2\kappa_2$ $\epsilon_2\lambda_2$
IgD	δ	None	κ or λ	$\delta_2\kappa_2$ $\delta_2\lambda_2$

V region

- 110 N terminal a.a of both L and H chains
- 3 Hypervariable regions** or complementarity determining regions CDR 1, 2 and 3 show greatest amount of variability: participate in Ag-Ab rxn
- Framework region:** the less variable stretches between CDRs
- The CDR of both L and H are brought together in the tertiary S. to make the binding site which is complementary to the epitope.

Ag-Ab RX

- Bonds between Ag and Ab

H-bonds

Hydrophobic bonds

Ionic bonds

Van der Waals forces

- Binding

-2 Abs with 2 specificities have different a.a sequences in the V

-2 Abs with different a.a sequences have specificity for the same epitope but have different affinities

-A specific Ab against one hapten may react with other Ag e.g DNP and sRBCs (redundancy).

Characteristics of hinge region

-Immunoglobulins contain hinge between

C_H1 & C_H2 . (with possible exception of IgM & IgE), The hinge region is replaced by an additional Ig domain (about 12 a.a)

-No homology between a.a sequence of hinge & heavy chains

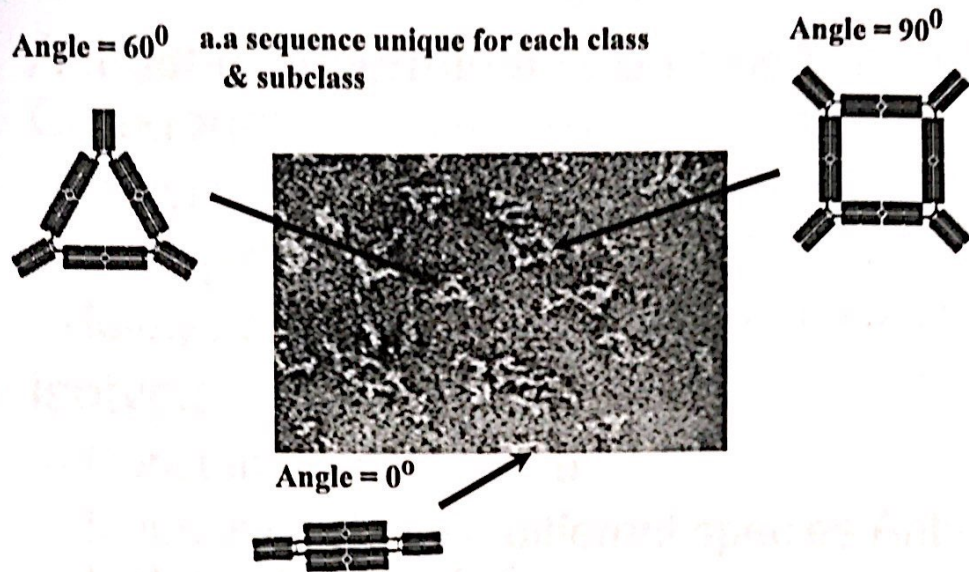
-a.a sequence differs with different classes

-Comprised of many *cysteine* and *proline* residues

Cysteine involved in formation of interchain disulfide bonds

-*Proline* prevents folding in a globular structure, allowing flexibility between two Fab arms of the Y-shaped antibody; allows open & close to accommodate binding to two epitopes.

Hinge region



Function of Fab region Binding the Ag

Functions of Fc region

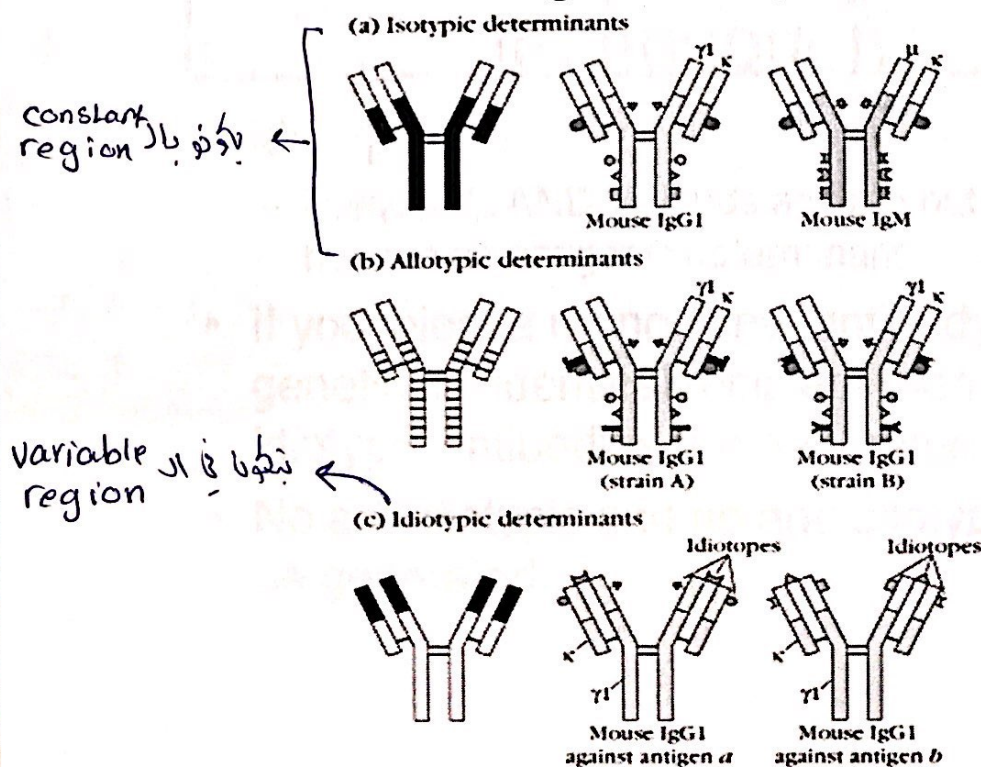
- Activation of complement system
- Opsonization : the phagocytes have FcR
- Cross placenta: bramble R for Fc

Antibodies Act As Immunogens

- Antigenic Determinants on Abs Fall in 3 Categories *idiotypes/anti-idiotypes*
 - Isotypic *idiotypes; if we inject IgG of a mouse into a rabbit it becomes an antigen for the ~~mouse~~ rabbit and it's called idio type*
 - Allotypic *and it's called idio type*
 - Idiotypic *anti-idiotypes; the antibodies made against the IgG of the mouse are called anti-idiotypes*
- Isotypic
 - Constant Region Of Ab
 - If you inject Ab in a different species Anti-Isotype is generated
 - within the same species, No Anti-isotype

there are some differences between one person and another in regions and that difference is 5%.

Antibodies are good antigens



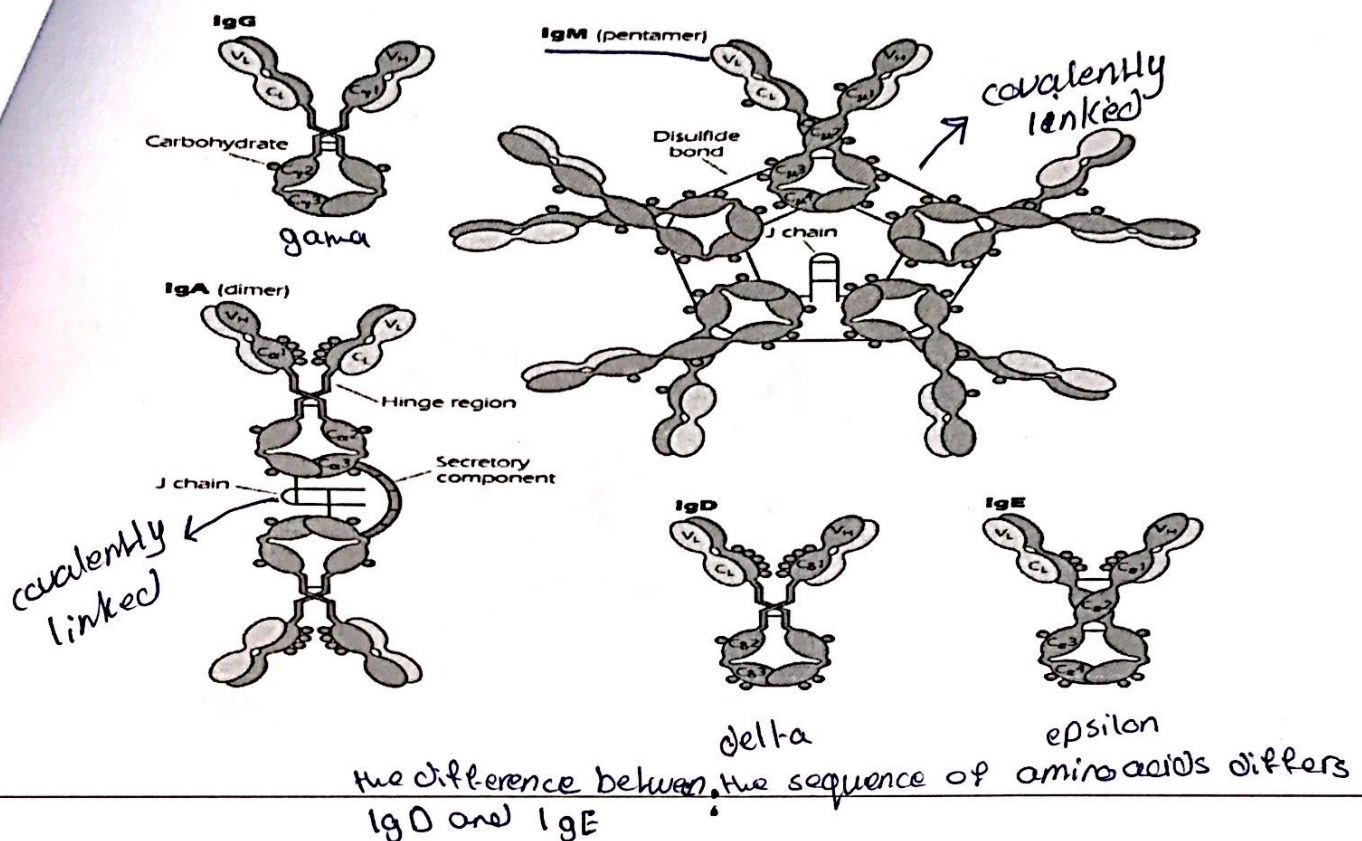
Antibodies Act As Immunogens

- Allotype
 - Even though same isotypes within one species small differences (1-4 a.a) arise in different individuals (form of polymorphism)
 - If injected with such Ab, you generate anti-allotype Ab
 - During pregnancy
 - Blood transfusion

Antibodies Act As Immunogens

- Idiotypic
 - Unique V_H AND V_L binds antigen but can also behave as antigenic determinant
- If you inject a monoclonal antibody into a genetically identical recipient then anti-idiotypic antibodies are generated
- No anti-isotypic and no anti-allotypic Abs will be generated

Ig classes



Biological Activities of IgG

the most abundant Ig in living is IgG

- Most abundant immunoglobulin 80% of serum Ig (~10mg/mL)
- Equal distribution between intravascular and extravascular spaces
blood between the cells
- Half life 23 days except IgG3 (7 days)
- Brambell R (FcR) a protection receptor bind the Fc region.
found in cellular endosomes, placenta, newborn intestine
- recycling of IgG after endocytosis of Ag-Ab
- saturable: at high conc.
- FCRP structurally related to MHC class I

فقط بعد الميلاد
بجدار الأم و
كان في دم بغير اللب
والبط

from breast milk

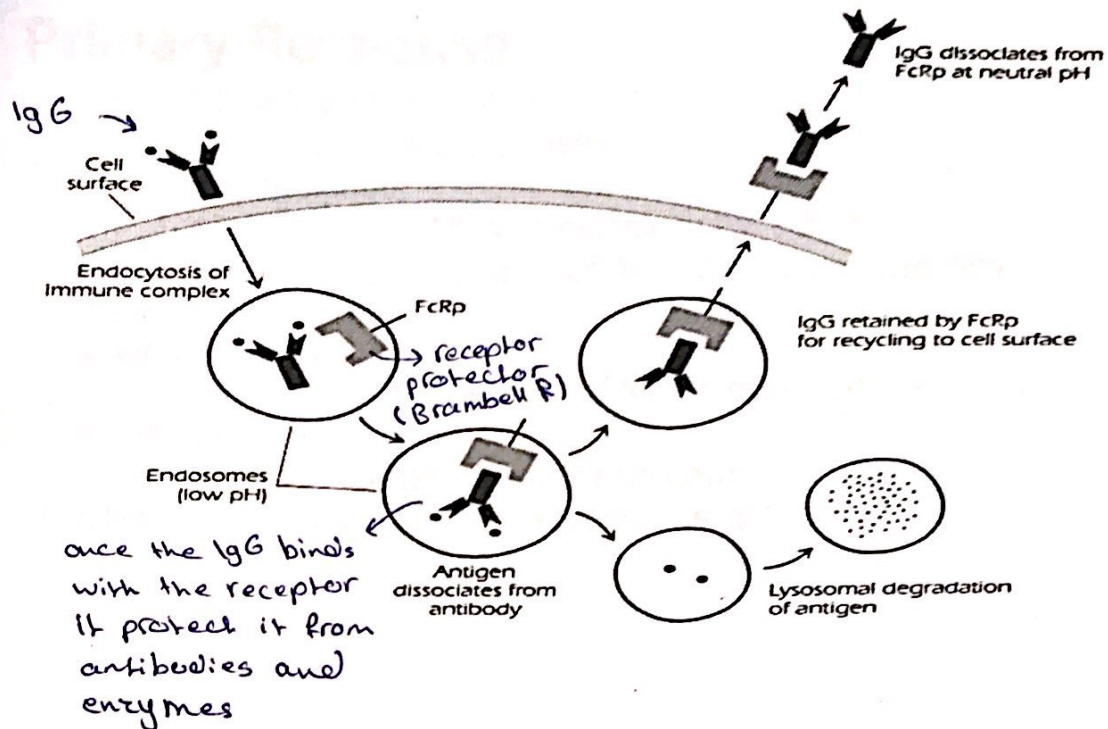
-Most suitable for passive immunization:

- . Longest half life
- . most abundant
- . Highly effective against m.o and toxins

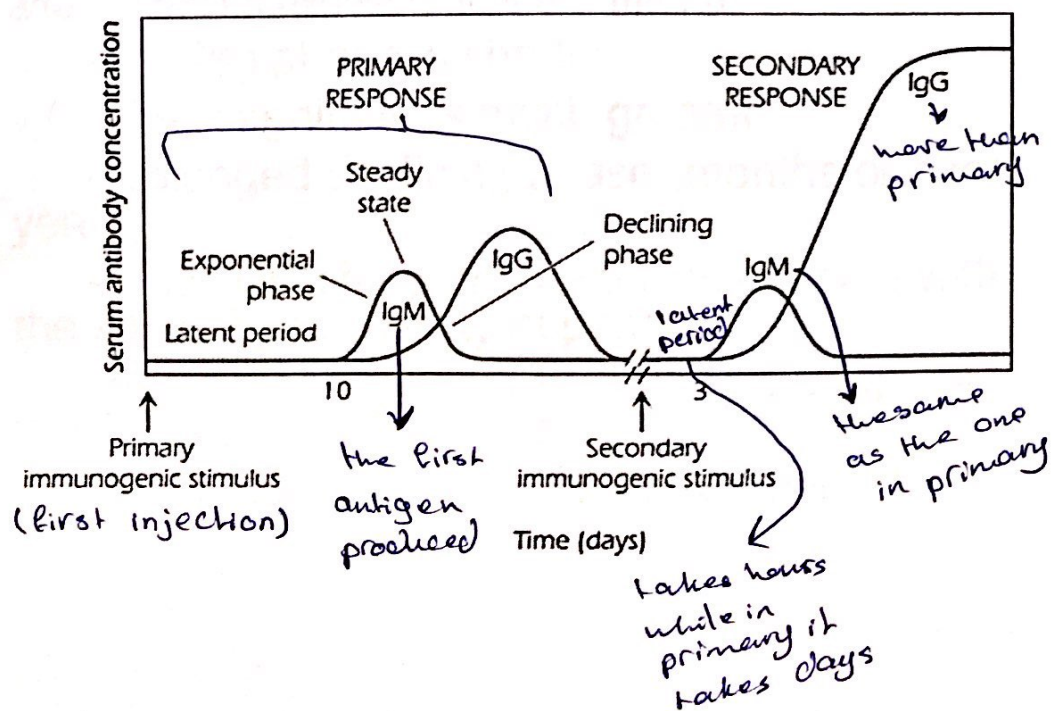
When an immediate reaction is required against toxins, the patient is given a ready antibodies intravenously. These antibodies are collected via injecting the toxin into an animal and taking its antibodies.

we use IgG in passive immunity because it has a long half-life

recycling of IgG after endocytosis of Ag-Ab



IgG Predominate in secondary i.r



Kinetics of Ab responses following immunization

Primary Response

the response to primary immunization.

Can be divided into several phases:

1. latent or lag period:

1-2 weeks before Ab is detectable in the serum

2. Exponential phase: the conc. Of Ab in the serum increases exponentially

3. steady state phase:

production and degradation of Ab are balanced

4. Declining phase

the immune system starts to shut down

The first Ab detected in primary responses is IgM

Secondary immune response

The immunized individual may be left with a pool of long-lived memory cells capable of mounting a secondary i.r. after a second exposure to the same Ag.

- The lag phase is shorter

- The magnitude is much greater

- prolonged declining phase (months or even years).

- Class switching: different classes of Ig with the same Ag specificity appear

- Affinity maturation: the binding strength of the Abs with the Ag increases

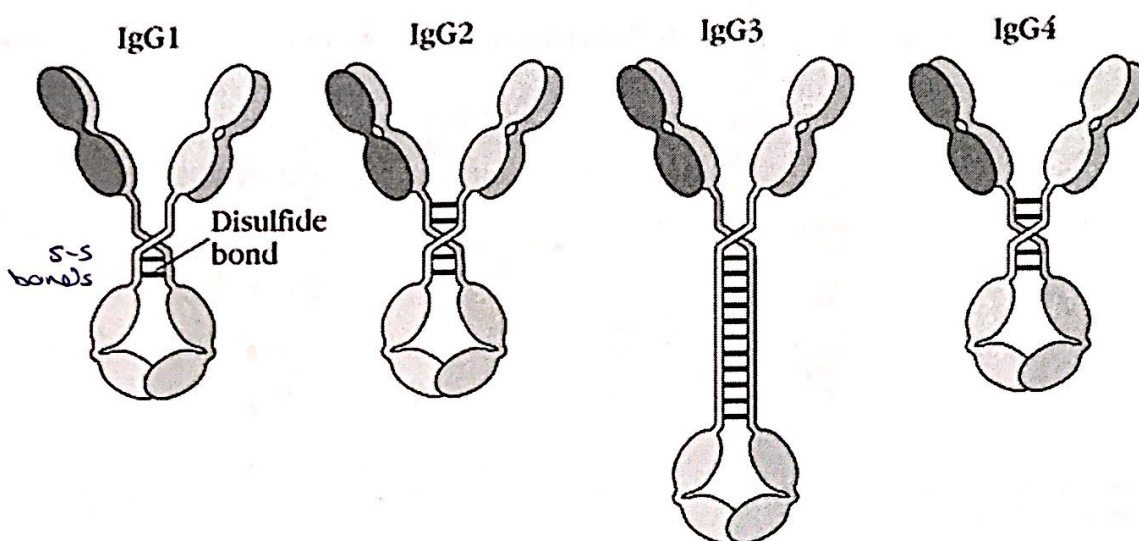
- IgG predominates but IgM also persist

Ig subclasses

- IgG1,2,3,4 (decreasing serum concentration)
- IgG2 does not cross placenta
- IgG4 does not activate complement
- IgG4 cannot bind phagocytes
- IgG1, 2 and IgG3 High affinity for FcR on phagocytic cells, good for opsonization

كيف نطعم اوجه الامتصاص في الـ IgG و انواعهم :
 بنطعم الـ IgA و الـ IgG كحشال و بنطعم الـ antibodies الـ IgG و بنطعم على
 الـ IgG فالعادة اني بتتربص فيها اوجه الامتصاص و اني بتفحص فيها اوجه الامتصاص

**IgG- most common in serum; monomeric
four subclasses**



Slight differences in structure; significant differences in function

• Passage through placenta

- IgG is the only Ig (except IgG2) cross placenta
- facilitated by expression of FcRp on placental cells
- 3-4 months of pregnancy: rapid increase in maternal IgG.
- During the 5th month the fetus begins to synthesize IgM and trace IgA.
- 4 months after birth, the infant begins to synthesize its own IgG
- Mediated by Fc region
- The FcRp is superexpressed in the intestinal tissue of neonates for 2 weeks
- Besides it confers immunity to infection, it is responsible for hemolytic disease of the newborn

Petus
if the ~~body~~ has high IgM it means that it has an infection.

IgM means that there's a ^{recent} ~~past~~ infection where IgG means that there's an infection بشكل عام

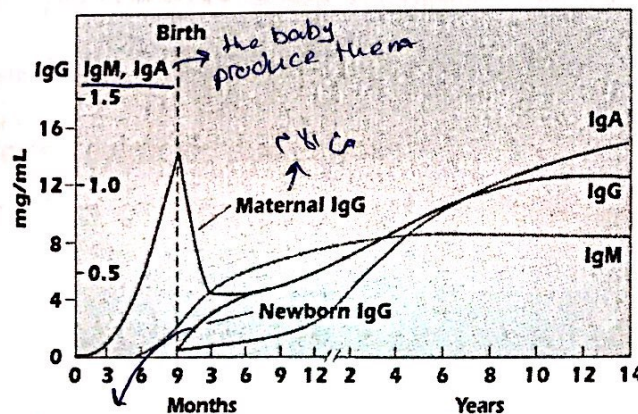


Figure 21.4. Concentration of immunoglobulins in the serum during human development.

بلد يصفه الطفل
Ig G

- **Agglutination:** IgG cause Clumping of particulate (insoluble) Ags.

Insoluble Ag-Ab complexes are easily phagocytosed and destroyed

- . **Precipitation:** reaction of soluble multivalent Ag with Ab

- . **Aggregation of IgG:**

precipitation with alcohol (purification of IgG)

heating at 56°C for 10 min (inactivate complement)

It is imperative that no aggregated IgG can be present in passively administered IgG:

- .aggregates still react with the Ag
- .attach to phagocytes
- .activate complement.
- .cause inflammation

- **Opsonization:** facilitation of phagocytosis due to presence of FcR on phagocytes

Pathogen-IgG-Fc....FcR-phagocyte

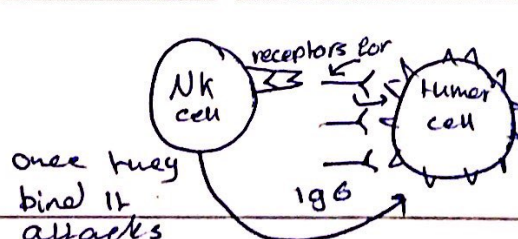
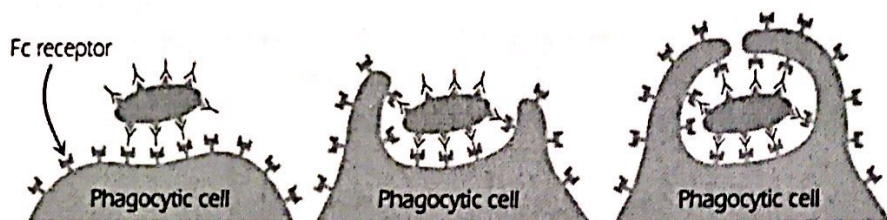
- **Ab-dependent Cell-mediated cytotoxicity**

NK cells-FcR.....FcR-IgG-Tumor cell or m.o

natural killer

NK release cytotoxic substances

- **Activation of Complement system:** IgG4 does not activate C



Neutralization of toxins and Viruses

- neutralization of tetanus and botulinus, snake venom

- Immobilization of Bac

IgG react with flagella or cilia arrest the movement of bac and prevent their spread

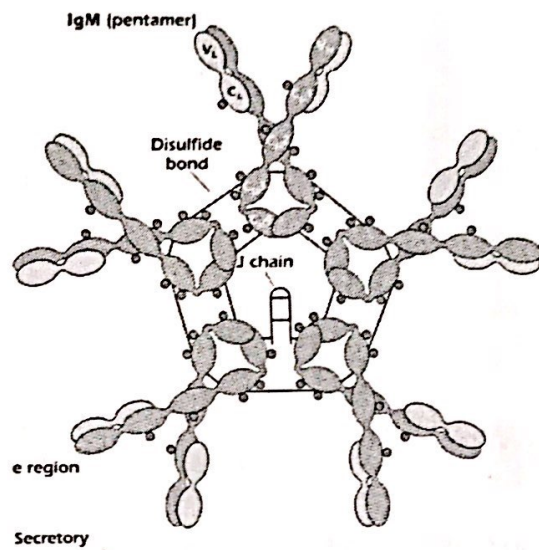
once the IgG bind to the viruse it can no longer divide

IgM

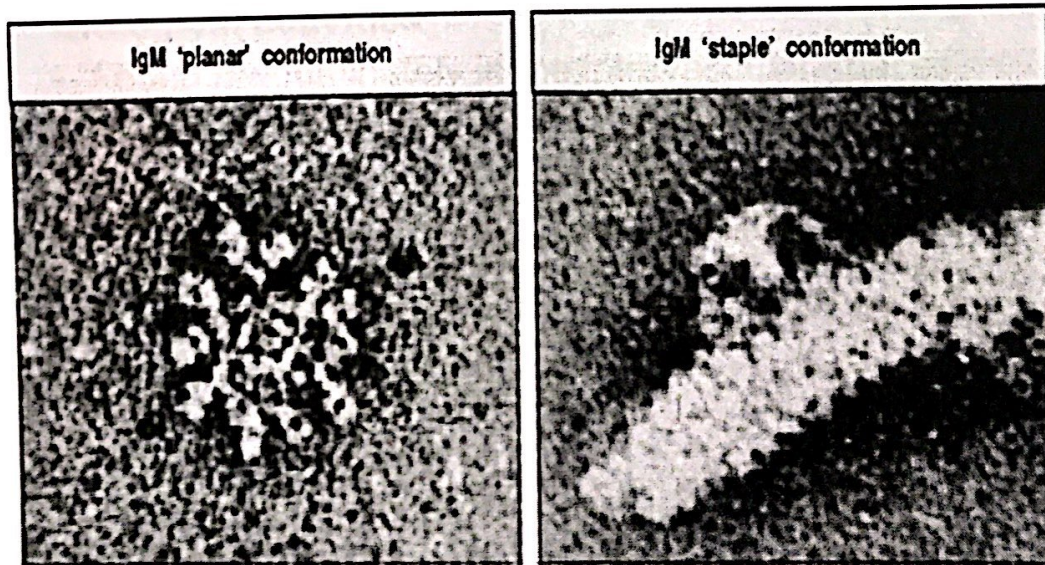
- **Structure**

- 2 types: **monomeric** or membrane IgM expressed on B-cells as BCR
- **Pentameric** IgM version is secreted
 - 5-10% of serum immunoglobulin (1.5mg/mL)
 - High M.W (900 KD) and 19S
 - Has an extra CH domain
 - High valence Ig (10 theoretical) but 5 due to steric hindrance.
 - The 5 monomers joined by a small chain (J chain) which is synthesized in the B cells. Mild treatment with reducing agents e.g mercaptoethanol break the J chain

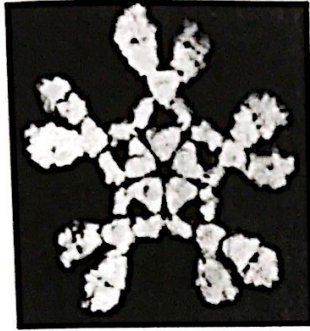
-**Monomeric IgM**: inserted in the B cell membrane



the other shape of IgM is IgM monomer (one molecule);
 IgM monomer only exist on the surface of B cells
 each B cell is different due to the Ig on its surface



Antigen-induced conformational changes in IgM

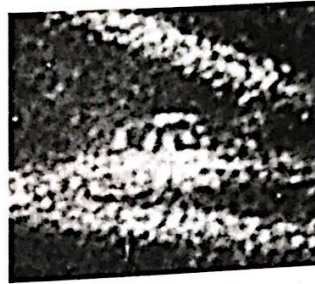


Planar or 'Starfish' conformation found in solution.

Does not fix complement



Staple or 'crab' conformation of IgM
Conformation change induced by binding to antigen.
Efficient at fixing complement



Biological activities

- Found predominately in the intravascular spaces.
- The half life is 5 days
- First Ig of primary immune response
 - provide an early line of defense against bacterial infection.
 - Elevated levels of IgM indicates either current or recent infection
- Unlike IgG, IgM do not cross placenta and the only Ig synthesized in the fetus, elevated level of IgM indicates congenital or perinatal infection.
- More efficient than IgG in complement activation: a single molecule is sufficient

- Due to their pentameric structure, IgM are very efficient in agglutination and suited to combat Ags with repetitive epitopes like LPS (T-indep. Ags).
- Include the **natural occurring Abs** against the RBCs Ags of the ABO system. So without known prior immunization, people with group O have isohemagglutinins of anti-A and Anti-b Ags.
- Cause transfusion reaction but does not cause hemolytic diseases of the newborn

IgA

- Structure
 - Monomeric IgA has no known function and a half life of 5.5 days and present mainly in serum
 - Dimeric or secreted sIgA:
 - Has 2 subclasses IgA1 (93%) and IgA2 (7%).
 - sIgA: dimer + J-chain polypeptide + secretory component (Poly IgR)

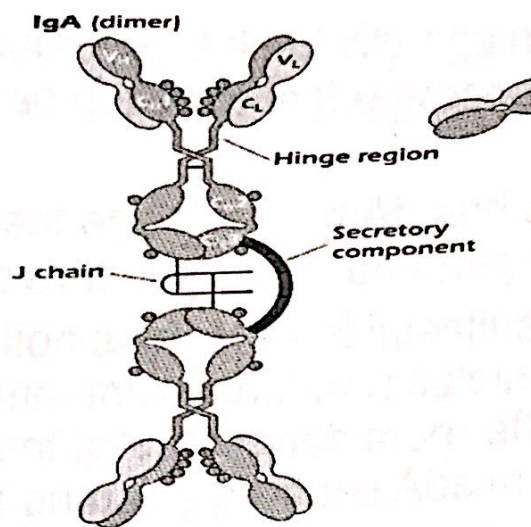


Figure 5.9 Structures of the five major

- Milk has more IgA than IgG specially the milk produced right after birth

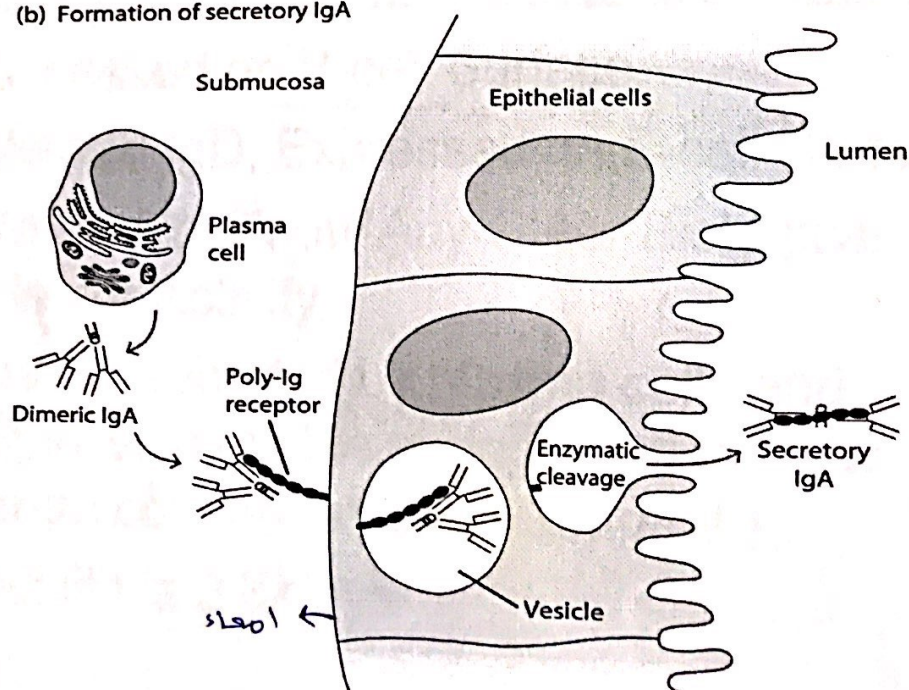
Biological activity

- Predominant Ig in secretions
 - Milk, saliva, tears, mucus
- The plasma cells synthesizing IgA molecules and the J chains to form the dimers. The plasma cells are located in the lamina propria that lies below the basement mem of many surface epithelia (e.g parotid gland, intestinal villi, tear glands, lactating breast, bronchial mucosa).
- The secretory component protect the sIgA from proteases. (It also transport pentameric IgM to mucosal surfaces in small amounts).

- IgA do not fix complement
- Possess bacteriicidal activity against gram negative bac but only in the presence of lysozyme.
- Is an efficient agglutinating Ab and against viruses prevent them from entering host cells.
- For protection against local intestinal infection, routes of immunization that result in local production of IgA are much more effective than routes that primarily produce Abs in serum.

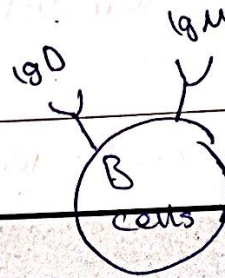
IgA Antibody Transport Across Cell (Transcytosis)

(b) Formation of secretory IgA



Antibody IgE (allergy)

- Very low serum concentration, $0.3\mu\text{g/mL}$ and the least half life (2 days).
- Participate in immediate hypersensitivities reactions. Ex. Asthma, anaphylaxis, hives
- Binds Mast Cells and Blood Basophils thru $\text{Fc}\epsilon\text{R}$
- Binding causes degranulation (Histamine Release)



IgD

- Expressed on B-cell Surface
- IgM and IgD, Expressed on B-cell Surface
- We Do Not Know Any Other Biological Effector Activity
- It is not secreted by plasma cells and highly susceptible to degradation (Low serum concentrations, $\sim 30\mu\text{g/mL}$).
- Half life is 2.8 days